José Federico Castro Tramontina

Software and Embedded Systems Developer | Cybersecurity Specialist

Applied Cryptography · Artificial Intelligence · Linux · Python · C/C++

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Buenos Aires | Mendoza, Argentina

Open to remote, hybrid, and on-site roles · Willing to relocate

PROFESSIONAL PROFILE

IT professional specializing in software development, embedded systems, applied cryptography, and cybersecurity. Experienced in designing and implementing real-time client-server solutions, process automation, systems administration, and industrial cybersecurity. Argentine Army officer with experience in team leadership, communications, and information security. Able to integrate software, hardware, networking, and security into real-world technology solutions.

TECHNICAL SKILLS

Programming languages: Python, C, C++, JavaScript, Bash, SQL, HTML5, CSS3.

Embedded systems and IoT: Arduino, ESP32, STM32, Raspberry Pi Pico, Raspberry Pi, KiCad, hardware-software integration.

Cybersecurity: Cryptography, post-quantum cryptography, OT/ICS security, ISO 27001, Modbus TCP/IP, OpenSSL, Wireshark, Nmap, tcpdump.

Scientific computing and AI: SageMath, Jupyter, MATLAB/GNU Octave, Antigravity, ChatGPT, Claude, Codex, DeepSeek, Gemini, llama.cpp, LM Studio, ComfyUI, OpenClaw.

Backend and data: FastAPI, WebSockets, SQLite, APIs, client-server architecture, Typer, Rich.

Industrial automation: PLCs, Ladder Logic, Structured Text, TIA Portal, EcoStruxure.

Systems, DevOps, and software quality: GNU/Linux, Docker, Podman, Git, GitHub, OpenSSH, uv, pytest, Hypothesis, Ruff, mypy, MkDocs, Samba, Crontab, VirtualBox.

PROFESSIONAL EXPERIENCE

Software and Embedded Systems Developer — Freelance

May 2026 – Present

- **Queue management system — Farmacia Vital Olivos:** design and implementation of a real-time client-server solution with a customer touchscreen kiosk, operator console, and multimedia queue display. Python, FastAPI, WebSockets, SQLite, HTML5, CSS3, JavaScript.

Cybersecurity Consultant — Freelance

November 2024 – Present

- **genomIT:** advisory services focused on regulatory compliance, personal data protection, and strengthening information security policies and controls aligned with ISO 27001.

IT Officer | Systems Administrator — Argentine Army

June 2014 – Present

- **Systems administration, communications, and security:** automation with Python, Bash, and Google Apps Script, Moodle platform maintenance, and configuration of computing and radio communications equipment. Responsible for communications and information security. Served as Section Leader, Echelon Leader, Communications Officer, and leader of teams of up to 40 personnel.

KEY PROJECTS

HK17.2: design, implementation, and cryptanalysis of a post-quantum key exchange protocol based on Cayley-Dickson algebras, including evaluation through cryptographic attacks.

CryptoLab: applied cryptography laboratory integrating mathematical foundations, transparent educational implementations, modern library-backed primitives, standardized post-quantum cryptography (ML-KEM, ML-DSA, and SLH-DSA), reproducible validation, algorithm comparison, and controlled cryptanalysis laboratories.

industrialPKI: Industrial PKI testbed for securing Modbus communications with mutual TLS, SCADA integration, PLC emulation, and Raspberry Pi deployment.

secureGate: secure access control system for entry to the Mechatronics Laboratory at the Army Engineering College.

Secure embedded systems network: design and implementation of an embedded device architecture protected through post-quantum cryptography.

IFPLab: autonomous local AI research agent for integer factorization, featuring automatic hypothesis generation, SageMath-based computational experimentation, and iterative closed-loop feedback.

EDUCATION

PhD Candidate in Computer Science — Universidad Abierta Interamericana. Dissertation: *Design, Development, and Cryptanalysis of a Key Exchange Protocol Based on Cayley-Dickson Algebras.*

MS Student in Data Networks — Universidad Tecnológica Nacional, Facultad Regional Mendoza. Thesis: *Design and Implementation of a Secure Embedded Systems Network Using Post-Quantum Cryptography.*

MS in Cyber Defense — Facultad de Ingeniería del Ejército.

MS in Cryptography and Network Security — Facultad de Ingeniería del Ejército.

BS in Operations Management — Facultad del Ejército, Colegio Militar de la Nación.

BS Student in Electronics Engineering — Facultad de Ingeniería del Ejército. Full-time scholarship student.

TEACHING AND RESEARCH

University lecturer in Cryptography. Researcher in applied cryptography, industrial cybersecurity, and embedded systems. Author and co-author of scientific publications, speaker at national conferences, and master's thesis examining committee member.

Languages: Spanish: Native · English: B2 (CEFR)